

Furstenberg's $\times 2, \times 3$ Conjecture: A Computational-Diophantine Approach via Baker Separation and Partition Entropy

John A. Janik
john.janik@gmail.com

February 2026

Abstract

We present large-scale computational evidence and structural analysis bearing on Furstenberg's 1967 conjecture that the only Borel probability measures on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ jointly invariant under $T_2: x \mapsto 2x$ and $T_3: x \mapsto 3x$ are Lebesgue measure and measures supported on finite orbits. Our approach transfers the Baker–cellError machinery developed for the Collatz conjecture to the Furstenberg setting, producing three categories of results.

Phase 1 (Orbit structure): We scan 4.55×10^8 primes up to 10^{10} , verifying that the density of primes p for which $\langle 2, 3 \rangle = (\mathbb{Z}/p\mathbb{Z})^*$ converges to the Euler product $\prod_\ell (1 - 1/(\ell^2(\ell - 1))) = 0.69750\dots$ to five significant figures. We also compute continued-fraction convergents of $\log_2 3$ and multi-scale cellError resonant fractions, confirming that the resonant count stays $O(1)$ at every scale (Baker separation).

Phase 2 (Entropy bridge): We compute the joint partition $\mathcal{P}_2^{(j)} \vee \mathcal{P}_3^{(k)}$ for $j \leq 24, k \leq 15$, finding that: (i) the entropy deficit $\log(\#\text{atoms}) - H$ is *bounded* at ~ 0.18 nats across all scales; (ii) the minimum atom width equals the Bézout floor $1/(2^j \cdot 3^k)$ exactly; (iii) a fundamental asymmetry holds: $\text{ord}_{2^j}(3) = 2^{j-2}$ (index 2) while $\text{ord}_{3^k}(2) = \varphi(3^k)$ (index 1, primitive root), both verified computationally to exponent 40. These results constrain the entropy bridge argument: T_3 -invariance forces approximate uniformity on $\mathcal{P}_2^{(j)}$ atoms, which together with Rudolph's theorem would close the conjecture.

Phase 2b (Spectral gap): We prove analytically and verify numerically that the transfer matrix of T_3 on $\mathcal{P}_2^{(j)}$ has *all* non-trivial eigenvalues with modulus exactly $1/3$, giving spectral gap $2/3$ independent of j . The proof uses a telescoping product identity: $\prod_{i=0}^{L-1} |1 + 2 \cos \theta_i| = 1$ for every orbit of $\times 3$ on $\mathbb{Z}/2^j\mathbb{Z}$. This is verified to $j = 30$ via Fourier orbit products (2^{30} elements) and to $j = 10$ via direct matrix multiplication (1024×1024 matrices, 4 orbit cycles each). The uniform spectral gap gives exponential mixing: $\|T_3^n p - \text{uniform}\|_2 \leq (1/3)^n \cdot \|p - \text{uniform}\|_2$.

Phase 2c (Joint constraints): We build the combined $T_2 + T_3$ constraint matrix across 2–3 consecutive dyadic scales (up to 9216×9216 at $j = 10$) and compute exact rank formulas via Gaussian elimination: $\text{rank} = 7N - 1$ (2-scale) or $17N - 1$ (3-scale), with T_2 -invariance killing exactly N_{finest} degrees of freedom. However, the atom-mass DOF at the coarsest scale remains $N/2 + 1$, showing that the linear constraint system alone does not force $\mathbf{T}p = p$. This identifies *positivity* ($w_{k,s} \geq 0$) as the essential non-linear ingredient for the entropy bridge, consistent with the structure of Rudolph's theorem.

1 Introduction

Furstenberg's conjecture [3] is one of the central open problems in ergodic number theory.

Conjecture 1 (Furstenberg, 1967). *The only Borel probability measures on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ simultaneously invariant under $T_2: x \mapsto 2x \pmod{1}$ and $T_3: x \mapsto 3x \pmod{1}$ are Lebesgue measure λ and measures supported on finite orbits (rationals with denominator coprime to 6).*

The state of the art includes:

- **Rudolph (1990):** If μ is ergodic, jointly T_2 - and T_3 -invariant, and $h_\mu(T_2) > 0$, then $\mu = \lambda$.
- **Host (1995):** Simplified Rudolph’s proof.
- **Einsiedler–Katok–Lindenstrauss (2006):** Any counterexample has Hausdorff dimension 0.
- **The zero-entropy case remains completely open.**

Our approach uses Baker’s theorem on linear forms in logarithms—the same effective Diophantine tool that drives the analysis of the Collatz conjecture via “cellError” separation—to obtain quantitative constraints on the joint partition structure of T_2 and T_3 . The goal is to show that Baker separation forces positive entropy for any non-atomic jointly invariant measure, reducing Furstenberg to Rudolph.

2 Mathematical Framework

2.1 The Fourier picture

Characters of $\mathbb{T}$ are $\chi_n(x) = e^{2\pi i n x}$ for $n \in \mathbb{Z}$. If μ is T_2 -invariant, then $\hat{\mu}(n) = \hat{\mu}(2n)$; if also T_3 -invariant, then $\hat{\mu}(n) = \hat{\mu}(3n)$. Thus $\hat{\mu}$ is constant on orbits of $\langle \times 2, \times 3 \rangle$ acting on $\mathbb{Z}$. For $n \neq 0$, the orbit $\mathcal{O}(n) = \{2^a 3^b n : a, b \geq 0\}$ is infinite by multiplicative independence of 2 and 3. The orbit tends to ∞ , so by Riemann–Lebesgue $\hat{\mu}(n) = 0$ if μ is absolutely continuous. For singular μ , Riemann–Lebesgue fails, and the question is whether the Fourier coefficients can consistently remain nonzero along the rapidly diverging orbit.

2.2 Baker separation on the torus

Baker’s theorem [1] gives a lower bound on linear forms in logarithms. In the notation of [5]:

$$|a \log 2 + b \log 3| > \frac{C}{\max(|a|, |b|)^\kappa} \quad \text{for } (a, b) \neq (0, 0),$$

with $\kappa = 4.125$ (Rhin’s effective value). Define the *cellError*:

$$\text{cellError}(a, b) = a - \log_2 3 \cdot b.$$

Baker gives $|\text{cellError}(a, b)| > C / \max(|a|, |b|)^\kappa$, ensuring that the 2-adic and 3-adic structures are always “transverse.”

On the $(N \times N)$ torus, a cell (a, b) is *resonant* if $|\text{cellError}(a, b)| < 1/N$. Baker bounds the resonant count at $O(1)$ (independent of N), so the resonant *fraction* decays as $O(1/N)$. This is the “safe cell density” of the Collatz framework, now applied to the pure $\times 2, \times 3$ action.

2.3 The entropy bridge conjecture

Conjecture 2 (Entropy Production). *If μ is a non-atomic, ergodic Borel probability measure on $\mathbb{T}$ that is simultaneously T_2 - and T_3 -invariant, then*

$$h_\mu(T_2) = \lim_{j \rightarrow \infty} \frac{1}{j} H_\mu \left(\bigvee_{i=0}^{j-1} T_2^{-i}(\mathcal{P}_2) \right) > 0,$$

where $\mathcal{P}_2 = \{[0, \frac{1}{2}), [\frac{1}{2}, 1)\}$.

If Conjecture 2 holds, then Rudolph’s theorem immediately implies $\mu = \lambda$, closing Furstenberg’s conjecture.

The partition $\mathcal{P}_2^{(j)} = \bigvee_{i=0}^{j-1} T_2^{-i}(\mathcal{P}_2)$ has atoms $\{[m/2^j, (m+1)/2^j) : 0 \leq m < 2^j\}$ with boundary points at the 2-adic rationals $\{m/2^j\}$. Similarly, $\mathcal{P}_3^{(k)}$ has boundaries at $\{m'/3^k\}$. The joint partition $\mathcal{P}_2^{(j)} \vee \mathcal{P}_3^{(k)}$ overlays both sets of boundaries. Baker’s theorem ensures that 2-adic and 3-adic boundaries never coincide (except at 0), and quantifies their minimal separation.

3 Computational Methods

All computations were performed on an Intel Core Ultra 9 275HX (24 cores, no hyperthreading) with 128 GB DDR5, using OpenMP parallelization.

3.1 Phase 1: Prime orbit scan (`furstenberg_orbits.c`)

For each prime p with $\gcd(p, 6) = 1$ up to $P_{\max}$:

1. Factor $p - 1$ by trial division using precomputed primes to 10^5 .
2. Compute $\text{ord}_p(2)$ and $\text{ord}_p(3)$ via the factorization of $p - 1$: start with $\text{ord} = p - 1$, then for each prime factor q of $p - 1$, divide out q while $x^{\text{ord}/q} \equiv 1 \pmod{p}$.
3. Compute $|\langle 2, 3 \rangle| = \text{lcm}(\text{ord}_p(2), \text{ord}_p(3))$ using the cyclic group identity.
4. Test full generation: $\langle 2, 3 \rangle = (\mathbb{Z}/p\mathbb{Z})^*$ iff $\text{lcm} = p - 1$.

The sieve and prime collection use bit-packed Eratosthenes (625 MB at 10^{10}). Checkpoints are saved every 200K primes via atomic rename. At 10^{10} , the program processes 7.9×10^6 primes/s on 24 threads.

3.2 Phase 2: Entropy computation (`furstenberg_entropy.c`)

For each (j, k) pair, the joint partition $\mathcal{P}_2^{(j)} \vee \mathcal{P}_3^{(k)}$ has boundary points $\{m/2^j\} \cup \{m'/3^k\}$. In the common denominator $D = 2^j \cdot 3^k$, these become integers:

$$\{m \cdot 3^k : 0 \leq m < 2^j\} \cup \{m' \cdot 2^j : 0 \leq m' < 3^k\}.$$

We sort these $2^j + 3^k - 1$ integers (only 0 is shared), compute the gaps between consecutive values, and derive:

- Atom count: $N_{\text{atoms}} = 2^j + 3^k - 1$.
- Minimum and maximum gap in integer units.
- Entropy: $H = -\sum_i w_i \ln w_i$ where $w_i = \text{gap}_i / D$.
- Entropy deficit: $\ln(N_{\text{atoms}}) - H$.

For $j \leq 24$, $k \leq 15$, the boundary array fits in ~ 250 MB.

Multiplicative orders $\text{ord}_{2^j}(3)$ and $\text{ord}_{3^k}(2)$ are computed by factoring $\varphi(2^j)$ and $\varphi(3^k)$ and iteratively dividing out prime factors while the power test passes, using 128-bit modular arithmetic.

3.3 Phase 2b: Spectral verification (furstenberg_spectrum.c)

The transfer matrix spectrum is verified in two ways: (1) Fourier orbit products over all orbits of $\times 3$ on $\mathbb{Z}/2^j\mathbb{Z}$ for $j \leq 20$, and representative orbits for $j \leq 30$; (2) direct matrix power iteration on the $2^j \times 2^j$ transfer matrix for $j \leq 10$, tracking the geometric mean decay over complete orbit cycles of length $L = \text{ord}_{2^j}(3) = 2^{j-2}$.

3.4 Phase 2c: Joint constraint analysis (furstenberg_lift.c)

For each scale j , we construct the combined $T_2 + T_3$ constraint matrix (a dense $9N \times 9N$ system for the 2-scale case, with $N = 2^j$) and compute its rank via Gaussian elimination with column partial pivoting and tolerance 10^{-10} . After row reduction, we project the null space onto subsets of variables (coarse sub-pieces, atom masses) and compute the projected null space dimension via a second rank computation. At $j = 10$, the matrix requires ~ 680 MB and the elimination takes ~ 10 s on a single core. For the 3-scale case (scales $j, j+1, j+2$), the system has $21N$ variables and $25N$ constraints; this is limited to $j \leq 8$ (~ 275 MB).

4 Phase 1 Results: Orbit Structure at Scale 10^{10}

4.1 Full generation density

We scanned 455,052,509 primes coprime to 6 up to 10^{10} . Table 1 shows the full generation percentage by decade.

Table 1: Full generation density $|\langle 2, 3 \rangle|/(p-1) = 1$ by decade.

Range	Primes	Full gen	Pct
$[5, 10^2)$	23	18	78.26%
$[10^2, 10^3)$	143	105	73.43%
$[10^3, 10^4)$	1,061	756	71.25%
$[10^4, 10^5)$	8,363	5,854	70.00%
$[10^5, 10^6)$	68,906	48,075	69.77%
$[10^6, 10^7)$	586,081	408,790	69.75%
$[10^7, 10^8)$	5,096,876	3,555,045	69.75%
$[10^8, 10^9)$	45,086,079	31,448,732	69.75%
$[10^9, 10^{10})$	404,204,977	281,933,832	69.75%
Total	455,052,509	317,401,207	69.75%

By Chebotarev’s density theorem, the theoretical density of primes where $\langle 2, 3 \rangle = (\mathbb{Z}/p\mathbb{Z})^*$ is given by the Euler product:

$$\delta = \prod_{\ell \text{ prime}} \left(1 - \frac{1}{\ell^2(\ell-1)} \right) = 0.69750136 \dots \quad (1)$$

The product converges rapidly: $\ell = 2$ contributes $3/4$; $\ell = 3$ contributes $17/18$; $\ell = 5$ contributes $99/100$. Our empirical 69.750% matches to five significant figures.

Interpretation. At “scale p ” (working on $\mathbb{Z}/p\mathbb{Z}$), the $\langle \times 2, \times 3 \rangle$ semigroup covers all of $(\mathbb{Z}/p\mathbb{Z})^*$ for $\sim 70\%$ of primes. For the remaining $\sim 30\%$, non-trivial invariant subsets exist, but these are *finite orbits*—precisely what Furstenberg allows. The conjecture is consistent at every finite prime scale.

4.2 Index distribution

For non-full-generation primes, the index $(p-1)/|\langle 2, 3 \rangle|$ characterizes the failure mode. Table 2 shows the distribution over all 455M primes.

Table 2: Index distribution over 4.55×10^8 primes.

Index	Count	Pct	Interpretation
1	317,401,207	69.75%	Full generation
2	93,355,317	20.52%	Both are QR mod p
3	17,982,555	3.95%	Both are cubic residues
4	9,334,173	2.05%	4th power residues
5	3,180,503	0.70%	
6	4,494,649	0.99%	Composite: 2×3
8	2,722,949	0.60%	
12	1,123,071	0.25%	Highly composite
24	327,576	0.07%	Highly composite

Highly composite indices (6, 12, 24) are over-represented because they have many prime-power divisors, each contributing a factor to the probability.

4.3 Maximum index growth

Table 3 shows the maximum index per decade.

Table 3: Maximum index $\max_p(p-1)/|\langle 2, 3 \rangle|$ by decade.

Decade	Max index
$[10^2, 10^3)$	10
$[10^3, 10^4)$	56
$[10^4, 10^5)$	72
$[10^5, 10^6)$	170
$[10^6, 10^7)$	519
$[10^7, 10^8)$	2,834
$[10^8, 10^9)$	3,235
$[10^9, 10^{10})$	35,590

The worst-case primes have very smooth $p-1$ (many small prime factors), causing both 2 and 3 to land in tiny subgroups. The minimum fraction $|\langle 2, 3 \rangle|/(p-1)$ in the last decade is 2.81×10^{-5} , meaning the generators cover only 0.003% of $(\mathbb{Z}/p\mathbb{Z})^*$.

4.4 CF convergents of $\log_2 3$

The continued fraction $\log_2 3 = [1; 1, 1, 2, 2, 3, 1, 5, 2, 23, \dots]$ gives the best rational approximations p_n/q_n to $\log_2 3$, equivalently the closest near-equalities $2^{p_n} \approx 3^{q_n}$. Table 4 shows selected convergents with their cellError and the ratio to Baker’s lower bound.

The err/Baker ratio grows rapidly: Baker’s lower bound is exponentially weaker than the actual CF approximation rate. This “gap” is the quantitative room in which the entropy bridge must operate.

Table 4: CF convergents of $\log_2 3$ and comparison to Baker’s bound.

n	a_n	p_n/q_n	$ \text{cellError} $	err/Baker
4	2	19/12	1.96×10^{-2}	6.6×10^3
8	2	1054/665	6.30×10^{-5}	1.8×10^{10}
13	1	301994/190537	9.31×10^{-8}	1.1×10^{20}
19	1	630138897/397573379	1.16×10^{-10}	1.4×10^{22}

4.5 Multi-scale cellError resonance

On the $N \times N$ torus with threshold $\delta = 1/N$, we count cells (a, b) with $|\text{cellError}(a, b)| < \delta$. Table 5 shows the results.

Table 5: CellError resonant fraction at scales $N = 6^k$.

k	$N = 6^k$	Resonant cells	Fraction
1	6	1	0.167
2	36	2	0.056
3	216	2	0.009
4	1,296	2	0.0015
5	7,776	3	0.00039
6	46,656	1	0.000021
8	1,679,616	6	0.0000036

The resonant count stays $O(1)$ at every scale, consistent with Baker separation (at most ~ 2 CF convergents in range at any scale). The resonant fraction decays as $\sim 1/N$, confirming that the torus becomes overwhelmingly “safe” at large scales.

5 Phase 2 Results: The Entropy Bridge

5.1 Multiplicative order asymmetry

Proposition 1 (Verified computationally to exponent 40). *For all $j \geq 3$:*

$$\text{ord}_{2^j}(3) = 2^{j-2}, \quad \text{index} [\varphi(2^j) : \text{ord}_{2^j}(3)] = 2.$$

For all $k \geq 1$:

$$\text{ord}_{3^k}(2) = \varphi(3^k) = 2 \cdot 3^{k-1}, \quad \text{index} = 1 \quad (2 \text{ is a primitive root mod } 3^k).$$

Both statements are classically known (the second is Artin’s conjecture restricted to $g = 2$, $p = 3$, verified for all prime powers 3^k). We verify them computationally through 128-bit modular exponentiation for all $j, k \leq 40$ (80 tests, all matching predictions).

This asymmetry is the structural foundation of the entropy bridge:

- T_3 (multiplication by 3) acts on $\mathcal{P}_2^{(j)}$ with period 2^{j-2} , generating only *half* of $(\mathbb{Z}/2^j\mathbb{Z})^*$.
- T_2 (multiplication by 2) acts on $\mathcal{P}_3^{(k)}$ with full period $\varphi(3^k)$, generating *all* of $(\mathbb{Z}/3^k\mathbb{Z})^*$.

The index-2 constraint on the 2-adic side means that T_3 -invariance forces a measure to be constant on orbits of size 2^{j-2} , but leaves one binary degree of freedom (the relative weight of the two cosets).

5.2 Joint partition atom analysis

For balanced scales $k(j) = \lfloor j \log 2 / \log 3 + 1/2 \rfloor$ (where $2^j \approx 3^k$), the joint partition $\mathcal{P}_2^{(j)} \vee \mathcal{P}_3^{(k)}$ has $N = 2^j + 3^k - 1$ atoms. Table 6 shows the entropy statistics.

Table 6: Entropy of $\mathcal{P}_2^{(j)} \vee \mathcal{P}_3^{(k)}$ under Lebesgue measure (balanced scales).

j	k	Atoms	H (nats)	H/j	Deficit	Efficiency
4	3	42	3.590	0.897	0.148	0.960
8	5	498	6.020	0.752	0.191	0.969
12	8	10,656	9.101	0.758	0.173	0.981
16	10	124,584	11.541	0.721	0.192	0.984
20	13	2,642,898	14.611	0.731	0.177	0.988
24	15	31,126,122	17.063	0.711	0.190	0.989

Key observation: the entropy deficit is bounded. Across all tested scales, the deficit $\ln(N) - H$ stays in the interval $[0.148, 0.192]$ nats. The efficiency $H/\ln(N)$ increases monotonically toward 1. Under Lebesgue measure, H/j converges to $\log 2 \approx 0.693$ from above (the excess comes from the additional information in the 3-adic refinement).

Proposition 2 (Bounded entropy deficit, empirical). *For all tested (j, k) with $j \leq 24$ and $k = k(j)$ balanced:*

$$\ln(2^j + 3^k - 1) - H_\lambda(\mathcal{P}_2^{(j)} \vee \mathcal{P}_3^{(k)}) \leq 0.20 \text{ nats},$$

where λ denotes Lebesgue measure.

This bounded deficit means the atom widths are approximately uniform at all scales. The few thin atoms (width $\sim 1/(2^j \cdot 3^k)$, achieving the Bézout floor) contribute negligibly to entropy.

5.3 Gap anatomy

The atoms of $\mathcal{P}_2^{(j)} \vee \mathcal{P}_3^{(k)}$ arise from interleaving two arithmetic progressions in $[0, D)$ with $D = 2^j \cdot 3^k$: the 2-adic boundaries $\{m \cdot 3^k\}$ (step 3^k) and the 3-adic boundaries $\{m' \cdot 2^j\}$ (step 2^j).

Proposition 3 (Bézout floor). *For all (j, k) tested (up to $j = 24$), the minimum gap between consecutive boundary points in the common-denominator representation is exactly 1. That is,*

$$\min_{\text{atoms}} \text{width} = \frac{1}{2^j \cdot 3^k}.$$

This minimum is achieved by the Bézout identity: there exist m_0, m'_0 with $m_0 \cdot 3^k - m'_0 \cdot 2^j = 1$.

Three types of gaps occur between consecutive boundary points:

Table 7: Gap types for $\mathcal{P}_2^{(12)} \vee \mathcal{P}_3^{(8)}$ ($D = 2^{12} \cdot 3^8 = 26,873,856$, $N = 10,656$ atoms).

Gap type	Count	Width	Meaning
3-adic $\rightarrow$ 3-adic	2,464	$2^{12} = 4096$	3-adic boundaries well-separated
Mixed ($2 \leftrightarrow 3$)	8,192	1–4096	Baker-governed
2-adic $\rightarrow$ 2-adic	0	—	Impossible ($3^8 > 2^{12}$)

The zero count of pure 2-adic gaps when $3^k > 2^j$ means that every pair of 2-adic boundaries has a 3-adic boundary between them—the 3-adic partition is “finer” at this scale. The mixed gaps, governed by Baker’s coprimality floor, have mean $\sim 3^k/2$ and minimum exactly 1.

5.4 Minimum atom width decay

The ratio of minimum to mean atom width decays exponentially:

Table 8: Minimum-to-mean atom width ratio (balanced scales).

j	min/mean	$\approx$
8	8.0×10^{-4}	$\sim 2^{-10}$
12	4.0×10^{-4}	$\sim 2^{-11.3}$
16	3.2×10^{-5}	$\sim 2^{-14.9}$
20	1.6×10^{-6}	$\sim 2^{-19.3}$
24	1.3×10^{-8}	$\sim 2^{-26.2}$

Despite the exponential shrinkage of the thinnest atom, the entropy deficit remains bounded because there is *exactly one* Bézout-thin atom per partition, and its entropy contribution $w \ln(1/w) \rightarrow 0$ as $w \rightarrow 0$.

5.5 Transfer matrix spectral gap

The transfer matrix of $T_3: x \mapsto 3x$ on the partition $\mathcal{P}_2^{(j)}$ is the $2^j \times 2^j$ doubly stochastic matrix $\mathbf{T}$ with entries:

$$T_{mi} = \begin{cases} 1/3 & \text{if } m \in \{3i, 3i+1, 3i+2\} \pmod{2^j}, \\ 0 & \text{otherwise.} \end{cases}$$

The column sums are 1 (each cell is hit by exactly 3 preimages from one source), and the row sums are 1 (each cell distributes equally to 3 targets).

Theorem 1 (Spectral gap of T_3 on $\mathcal{P}_2^{(j)}$). *For $j \geq 3$, all non-trivial eigenvalues of $\mathbf{T}$ have modulus exactly $1/3$. The spectral gap is $1 - 1/3 = 2/3$, independent of j .*

Proof sketch. In the Fourier basis $v_k(m) = \omega^{mk}$ with $\omega = e^{2\pi i/2^j}$, the matrix $\mathbf{T}$ maps $v_k \mapsto \lambda(k) \cdot v_{ks}$ where $s = 3^{-1} \pmod{2^j}$ and

$$\lambda(k) = \frac{1}{3} (1 + 2 \cos(2\pi ks/2^j)).$$

For an orbit $\{k_0, 3k_0, 9k_0, \dots\}$ of size L under $\times 3 \pmod{2^j}$, the eigenvalue modulus on the associated L -dimensional block is $(\prod_{i=0}^{L-1} |\lambda(k_i)|)^{1/L}$.

Using the identity $1 + 2 \cos \theta = (e^{3i\theta} - 1)/(e^{i\theta} - 1)$, the product telescopes:

$$\prod_{i=0}^{L-1} |1 + 2 \cos(2\pi k_i s/2^j)| = \frac{|e^{2\pi i \cdot 3^L k_0 s/2^j} - 1|}{|e^{2\pi i \cdot k_0 s/2^j} - 1|} = 1,$$

since $3^L \equiv 1 \pmod{2^j / \gcd(k_0, 2^j)}$. Therefore $|\text{eigenvalue}| = (1/3^L)^{1/L} = 1/3$. $\square$

Numerical verification. We verify this theorem in two independent ways:

1. *Fourier orbit products* ($j = 3$ to 30): For each $j \leq 20$, we enumerate *all* orbits of $\times 3$ on $\mathbb{Z}/2^j\mathbb{Z}$ and compute $\prod |1 + 2 \cos \theta_i|$; every orbit gives product = 1.0000000000 (to 10 decimal places). For $j = 21$ to 30 , we check representative orbits of $k = 1$ and $k = 2$; the product stays within 10^{-7} of 1 (double-precision limit for sums over 10^8 terms).
2. *Direct matrix multiplication* ($j = 3$ to 10): We build the full transfer matrix, apply it to an initial vector orthogonal to uniform, and track the geometric mean of the per-step decay ratio over complete orbit cycles of length $L = \text{ord}_{2^j}(3)$. Every cycle gives geometric mean = 0.3333333333 to 10 decimal places.

Structural observations.

- The number of orbits is $2j + 1$ (for $j \leq 20$, verified), reflecting the $j + 1$ powers of 2 dividing $\gcd(k, 2^j)$ plus the coset structure.
- The maximum orbit size is $2^{j-2} = \text{ord}_{2^j}(3)$, confirming Proposition 1.
- The eigenvalue $\lambda(2^{j-1}) = -1/3$ (the unique fixed point $k = 2^{j-1}$ under $\times 3$, since $3 \cdot 2^{j-1} \equiv 2^{j-1} \pmod{2^j}$).

Proposition 4 (Exponential mixing). *For any probability distribution p on $\mathcal{P}_2^{(j)}$ atoms:*

$$\|\mathbf{T}^n p - \text{uniform}\|_2 \leq (1/3)^n \cdot \|p - \text{uniform}\|_2.$$

After n applications of T_3 , the distribution is exponentially close to uniform, at rate $1/3$ per step.

This means T_3 -invariance at the measure level ($\mathbf{T}p = p$) forces $p = \text{uniform}$ on $\mathcal{P}_2^{(j)}$ atoms, which gives $H_\mu(\mathcal{P}_2^{(j)}) = j \log 2$. The remaining question is whether T_3 -invariance of a measure μ on $\mathbb{T}$ implies T_3 -invariance of the induced distribution on $\mathcal{P}_2^{(j)}$ atoms—i.e., whether the “lift” from partition to measure preserves the spectral gap constraint.

5.6 The lift problem: straddle analysis

The preimage $T_3^{-1}(I_m)$ of atom $I_m = [m/2^j, (m+1)/2^j)$ consists of three intervals of width $1/(3 \cdot 2^j)$:

$$T_3^{-1}(I_m) = \bigcup_{r=0}^2 \left[\frac{m + r \cdot 2^j}{3 \cdot 2^j}, \frac{m + r \cdot 2^j + 1}{3 \cdot 2^j} \right).$$

In the common denominator $D = 3 \cdot 2^j$, each preimage piece occupies the interval $[a, a+1)$ where $a = m + r \cdot 2^j$. Since atoms in D -coordinates span $[3k, 3k+3)$ and $a+1 \leq 3k+3$ (as $a \bmod 3 \leq 2$), every piece lies entirely inside one atom. No straddling occurs: the “straddle matrix” $\mathbf{S}$ equals the transfer matrix $\mathbf{T}$ exactly.

Proposition 5 (Degrees of freedom). *At each scale j , T_3 -invariance of μ imposes $N - 1$ independent linear constraints on the $3N$ sub-piece masses $\{w_{k,s}\}$ (where k indexes atoms and $s \in \{0, 1, 2\}$ indexes sub-pieces). Together with the N atom-sum constraints $w_{k,0} + w_{k,1} + w_{k,2} = p_k$, the system has rank $2N - 1$, leaving $N + 1$ degrees of freedom (or 2 per atom plus normalization).*

This is verified computationally for $j = 3$ to 12: the constraint rank is always $N - 1$, giving exactly 2 DOF per atom at every scale.

Interpretation. T_3 -invariance at a single scale j constrains the atom masses p_m but *not* the within-atom distribution. Each atom has 3 sub-pieces (one for each preimage branch), and the relative weights of these sub-pieces are free parameters. The transfer matrix equation $\mathbf{T}p = p$ is the *special case* where each sub-piece carries exactly $1/3$ of its atom’s mass.

Multi-scale constraints. At scale $j + 1$, each atom of $\mathcal{P}_2^{(j)}$ splits into two atoms of $\mathcal{P}_2^{(j+1)}$. Each old sub-piece (width $1/(3 \cdot 2^j)$) spans $\sim 4/3$ new atoms, so the scale- $(j + 1)$ constraints partially fix the within-atom distribution at scale j . After $\lceil \log_2 3 \rceil + 1 = 3$ refinements, each old sub-piece is resolved into separate atoms, and the scale- $(j + 3)$ spectral gap forces the resolved masses toward uniformity.

The question is: do *multi-scale* constraints (T_3 -invariance at scales j and $j+1$ simultaneously, plus T_2 -invariance linking the scales) eliminate the within-atom DOFs? This is the subject of the next section.

5.7 Joint $T_2 + T_3$ constraint rank analysis

We now build the full constraint system for joint T_2 - and T_3 -invariance across multiple dyadic scales, and compute its rank to determine the degrees of freedom exactly.

Mathematical setup (2-scale system)

Consider two consecutive scales j and $j + 1$, with $N = 2^j$. The sub-piece masses are:

$$\begin{aligned} \text{Scale } j: & \quad w_{k,s} \text{ for } k \in [0, N), \quad s \in \{0, 1, 2\} & (3N \text{ variables}), \\ \text{Scale } j+1: & \quad w'_{m,s} \text{ for } m \in [0, 2N), \quad s \in \{0, 1, 2\} & (6N \text{ variables}). \end{aligned}$$

The total is $9N$ unknowns. Four blocks of homogeneous linear constraints relate them:

1. **T_3 -invariance at scale j** (N equations): for each atom m ,

$$\sum_{s=0}^2 w_{m,s} = \sum_{r=0}^2 w_{\lfloor (m+rN)/3 \rfloor, (m+rN) \bmod 3}.$$

2. **T_3 -invariance at scale $j+1$** ($2N$ equations): same structure with $2N$ atoms and $6N$ fine variables.
3. **Refinement** ($3N$ equations): each coarse sub-piece is the union of two fine sub-pieces. Explicitly:

$$\begin{aligned} w_{k,0} &= w'_{2k,0} + w'_{2k,1}, \\ w_{k,1} &= w'_{2k,2} + w'_{2k+1,0}, \\ w_{k,2} &= w'_{2k+1,1} + w'_{2k+1,2}. \end{aligned}$$

These follow from the geometric subdivision: sub-piece (k, s) at scale j covers $[(3k+s)/(3 \cdot 2^j), (3k+s+1)/(3 \cdot 2^j))$, which at scale $j+1$ (in coordinates with denominator $3 \cdot 2^{j+1}$) spans $[6k+2s, 6k+2s+2)$ and splits at $6k+2s+1$.

4. **T_2 -linking** ($3N$ equations): T_2 -invariance relates sub-pieces across scales via

$$w_{k,s} = w'_{k,s} + w'_{k+N,s} \quad (0 \leq k < N, \quad s \in \{0, 1, 2\}),$$

since T_2^{-1} of a sub-piece at scale j decomposes into two sub-pieces at scale $j+1$ separated by N .

Rank computation

We compute the rank of the constraint matrix via Gaussian elimination. Table 9 shows the results for $j = 3$ to 10.

Proposition 6 (Joint constraint rank, 2-scale). *For the 2-scale system at scales $j, j+1$ with $N = 2^j$ ($j \geq 3$):*

$$\begin{aligned} \text{rank}(\text{no } T_2) &= 5N - 1, & \text{DOF}(\text{no } T_2) &= 4N + 1, \\ \text{rank}(\text{with } T_2) &= 7N - 1, & \text{DOF}(\text{with } T_2) &= 2N + 1. \end{aligned}$$

T_2 -invariance kills exactly $2N$ degrees of freedom.

Table 9: 2-scale constraint rank analysis ($9N$ variables, verified for $j = 3, \dots, 10$).

j	N	$9N$	DOF (no T_2)	DOF ($+T_2$)	T_2 kills	Atom DOF
3	8	72	33	17	16	5
4	16	144	65	33	32	9
5	32	288	129	65	64	17
6	64	576	257	129	128	33
7	128	1,152	513	257	256	65
8	256	2,304	1,025	513	512	129
9	512	4,608	2,049	1,025	1,024	257
10	1,024	9,216	4,097	2,049	2,048	513

Table 10: 3-scale constraint rank analysis ($21N$ variables, verified for $j = 3, \dots, 8$).

j	N	$21N$	DOF (no T_2)	DOF ($+T_2$)	T_2 kills
3	8	168	65	33	32
4	16	336	129	65	64
5	32	672	257	129	128
6	64	1,344	513	257	256
7	128	2,688	1,025	513	512
8	256	5,376	2,049	1,025	1,024

3-scale extension

The same analysis extends to three consecutive scales ($j, j+1, j+2$) with $21N$ variables and $25N$ constraint rows. Table 10 shows the results.

The 3-scale rank formulas are:

$$\begin{aligned} \text{rank}(\text{no } T_2) &= 13N - 1, & \text{DOF}(\text{no } T_2) &= 8N + 1, \\ \text{rank}(\text{with } T_2) &= 17N - 1, & \text{DOF}(\text{with } T_2) &= 4N + 1. \end{aligned}$$

In both cases, the total DOF follows the universal formula

$$\text{DOF}(\text{with } T_2) = N_{\text{finest}} + 1, \quad (2)$$

where N_{finest} is the number of atoms at the finest scale ($2N$ for 2-scale, $4N$ for 3-scale). The extra DOF beyond 1 (overall scaling) resides at the finest scale, which has no child constraints.

Projection analysis

To understand where the freedom lives, we project the null space of the full 2-scale system (with T_2) onto subsets of variables. Table 11 shows the projected null space dimension at each level.

Table 11: Projected null space dimension for the 2-scale system (pattern exact for $j = 3, \dots, 10$).

Projection	Null dim	Formula
Full system ($9N$ variables)	$2N + 1$	$N_{\text{finest}} + 1$
Coarse sub-pieces ($3N$ variables)	$N + 1$	
Coarse atom masses (N variables)	$N/2 + 1$	

The atom-mass DOF at the coarsest scale is $N/2 + 1$, and this value is *unchanged* between the 2-scale and 3-scale systems (verified for $j = 3, \dots, 8$). Adding more scales does not further constrain the coarsest-scale atom masses within the linear system.

Interpretation

The atom-mass $\text{DOF} = N/2 + 1 \gg 2$ means the *linear* constraint system does not force $\mathbf{T}p = p$ (which would give $\text{DOF} = 2$, corresponding to Lebesgue and δ_0). The extra $N/2 - 1$ dimensions correspond to signed “pseudo-measures” that satisfy all the T_3 , T_2 , and refinement constraints algebraically but violate positivity ($w_{k,s} \geq 0$).

This identifies **positivity** as the essential ingredient for the entropy bridge. The linear-algebraic structure establishes:

- T_2 -invariance halves the DOF at every scale (killing exactly N_{finest} directions);
- The spectral gap (Theorem 1) forces any *genuine* solution of $\mathbf{T}p = p$ to be uniform;
- But the measure-level T_3 -invariance constraint is weaker than $\mathbf{T}p = p$, leaving $\sim N/2$ directions of freedom in atom masses that are incompatible with positivity but satisfy the linear system.

Both Lebesgue measure (uniform: $w_{k,s} = 1/(3N)$ at scale j , $w'_{m,s} = 1/(6N)$ at scale $j + 1$) and δ_0 (point mass: $w_{0,0} = 1$, $w'_{0,0} = 1$, all others 0) are verified to be exact solutions of the full constraint system. These span a 2-dimensional subspace within the $(2N + 1)$ -dimensional null space. The remaining $2N - 1$ null directions have negative components and therefore cannot represent probability measures.

6 Synthesis: Architecture of the Entropy Bridge

The Phase 1 and Phase 2 data reveal the following architecture for the entropy bridge argument:

1. **Baker coprimality floor.** The minimum distance between a 2-adic and a 3-adic boundary is $1/(2^j \cdot 3^k)$, achieved exactly by Bézout. Atoms never collapse.
2. **Bounded entropy deficit.** $\ln(N) - H \leq 0.20$ nats for all tested balanced scales up to $j = 24$. The atom widths are approximately uniform under Lebesgue measure, with the non-uniformity entirely explained by a single Bézout-thin atom.
3. **Order asymmetry.** $\text{ord}_{2^j}(3) = 2^{j-2}$ (index 2), while $\text{ord}_{3^k}(2) = \varphi(3^k)$ (index 1). Any T_3 -invariant measure on $\mathcal{P}_2^{(j)}$ must be constant on orbits of size 2^{j-2} , leaving only two cosets as free parameters.
4. **Full generation at prime scales.** For 69.75% of primes p , $\langle 2, 3 \rangle = (\mathbb{Z}/p\mathbb{Z})^*$, leaving no room for non-trivial invariant measures. For the remaining primes, invariant subsets exist but are *finite orbits*—exactly what Furstenberg allows.
5. **Spectral gap** $= 2/3$. The transfer matrix of T_3 on $\mathcal{P}_2^{(j)}$ has all non-trivial eigenvalues with modulus exactly $1/3$ (Theorem 1). This gives exponential mixing at rate $1/3$ per step, so T_3 -invariance at the partition level forces exact uniformity.
6. **Joint constraint rank** $= 7N - 1$. The combined $T_2 + T_3$ constraint system at two consecutive scales has rank $7N - 1$ out of $9N$ variables (Proposition 6), leaving $\text{DOF} = 2N + 1$. T_2 -invariance kills exactly $2N$ directions, halving the freedom at every scale. However, the *linear* system does not force the transfer equation $\mathbf{T}p = p$: the atom-mass DOF is $N/2 + 1$, not 2. The extra directions are signed pseudo-measures that violate positivity.

6.1 The gap to close

Points (1)–(5) together show that:

- Baker separation ensures atoms never collapse (point 1).
- The entropy deficit is bounded under Lebesgue measure (point 2).
- T_3 -invariance at the partition level forces exact uniformity on $\mathcal{P}_2^{(j)}$ atoms via the spectral gap (points 3, 5).

The remaining question is whether T_3 -invariance of a *measure* μ implies T_3 -invariance of the induced distribution on partition atoms. Specifically, if μ is T_3 -invariant, does the atom mass vector $p_m = \mu([m/2^j, (m+1)/2^j))$ satisfy $\mathbf{T}p = p$?

If μ is T_3 -invariant and the partition $\mathcal{P}_2^{(j)}$ were T_3 -invariant (i.e., $T_3^{-1}(\mathcal{P}_2^{(j)}) \preceq \mathcal{P}_2^{(j)}$), then yes. But $\mathcal{P}_2^{(j)}$ is a T_2 -partition, not a T_3 -partition, so the refinement $T_3^{-1}(\mathcal{P}_2^{(j)})$ generally creates atoms that straddle the 2-adic boundaries.

Point (6) sharpens this obstacle. The joint $T_2 + T_3$ constraint system, even across multiple scales, leaves $N/2 + 1$ degrees of freedom in the atom masses—far more than the 2 needed for $\mathbf{T}p = p$. The extra directions are signed pseudo-measures that satisfy all linear constraints but violate positivity ($w_{k,s} \geq 0$). Adding more scales does not reduce this freedom: the atom-mass DOF is $N/2 + 1$ for both 2-scale and 3-scale systems.

This identifies **positivity** as the essential non-linear ingredient. The linear-algebraic framework determines the constraint architecture (exact ranks, DOF formulas), but the final step—showing that the only *non-negative* solutions are Lebesgue and δ_0 —requires a convexity or extremal-measure argument that goes beyond rank computation. One promising route is to exploit the spectral gap: since $\mathbf{T}$ contracts non-uniform components by factor $1/3$, any non-uniform positive eigenvector of the multi-scale refinement must amplify signed components at finer scales, eventually violating positivity.

7 Comparison with Prior Work

Our computational-Diophantine approach differs from the existing literature in several respects:

- **Rudolph (1990)** proves the conjecture under $h_\mu > 0$ using symbolic dynamics and Rohklin partitions. We aim to show that Baker’s effective bounds *force* $h_\mu > 0$, reducing to Rudolph rather than replacing it.
- **EKL (2006)** use measure rigidity on homogeneous spaces to show any counterexample has dimension 0. Our dimension argument (Phase 3, future work) would recover this result by an entirely different route: multi-scale Baker separation on the torus.
- **No prior work** has combined Baker’s theorem with explicit partition entropy computation. The bounded entropy deficit (Proposition 2) and the order asymmetry (Proposition 1) are, to our knowledge, new observations in this context.

8 Honest Assessment

Difficulty: Extreme. Furstenberg’s conjecture has resisted all approaches for over 55 years. The gap between “Baker separation makes resonances sparse” (which we have demonstrated) and “Baker separation forces entropy production” (which we conjecture) is precisely the gap that separates the known results (Rudolph, EKL) from the full conjecture.

What we have accomplished:

- Verified the Euler product prediction to five significant figures over 4.55×10^8 primes.
- Demonstrated bounded entropy deficit across six orders of magnitude in partition scale.
- Established the order asymmetry as a structural invariant of the $\times 2, \times 3$ system.
- Proved and verified the spectral gap $= 2/3$ theorem (Theorem 1), showing exponential mixing of T_3 on $\mathcal{P}_2^{(j)}$.
- Computed exact rank formulas for the joint $T_2 + T_3$ multi-scale constraint system (Proposition 6), identifying positivity as the essential non-linear ingredient for the entropy bridge.
- Built a computational framework (OpenMP, checkpointed, extensible to GPU) for further investigation.

Realistic best case: We prove the Entropy Production Conjecture, reducing Furstenberg to Rudolph. This would be a major theorem.

Realistic worst case: We produce a partial result (e.g., Furstenberg for measures satisfying an explicit Diophantine condition on their support), publishable independently, that clarifies the remaining obstacle.

References

- [1] A. Baker, *Linear forms in the logarithms of algebraic numbers*, Mathematika **13** (1966), 204–216.
- [2] M. Einsiedler, A. Katok, and E. Lindenstrauss, *Invariant measures and the set of exceptions to Littlewood’s conjecture*, Ann. of Math. **164** (2006), 513–560.
- [3] H. Furstenberg, *Disjointness in ergodic theory, minimal sets, and a problem in Diophantine approximation*, Math. Systems Theory **1** (1967), 1–49.
- [4] B. Host, *Nombres normaux, entropie, translations*, Israel J. Math. **91** (1995), 75–83.
- [5] G. Rhin, *Approximants de Padé et mesures effectives d’irrationalité*, Séminaire de Théorie des Nombres, Paris 1985–86, Progr. Math. **71**, Birkhäuser, 1987, pp. 155–164.
- [6] D. J. Rudolph, *$\times 2$ and $\times 3$ invariant measures and entropy*, Ergodic Theory Dynam. Systems **10** (1990), 395–406.